School of Engineering

REPUBLIC POLYTECHNIC

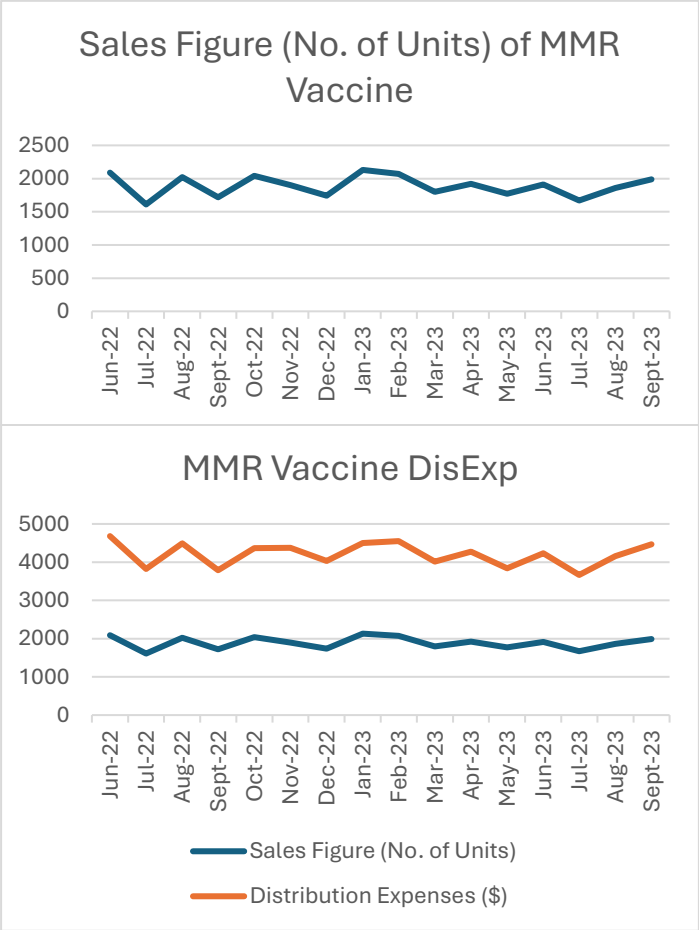
E211 – Operations Planning II

Worksheet for Lesson 26: Demand Forecasting

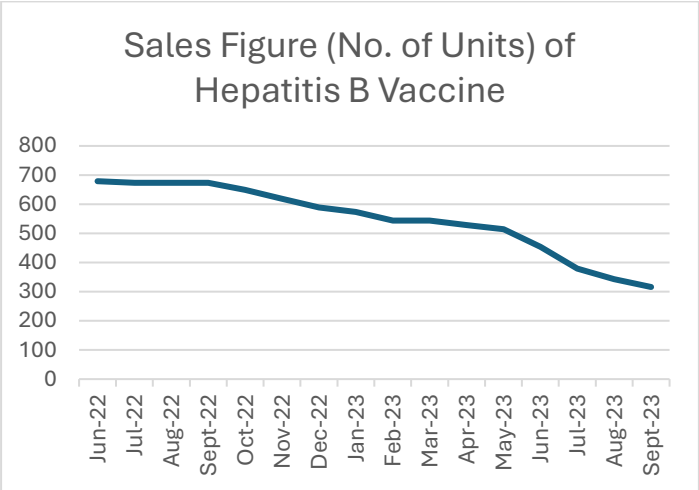
AY 2025 Sem 2 L26 Forecasting Data.xlsx

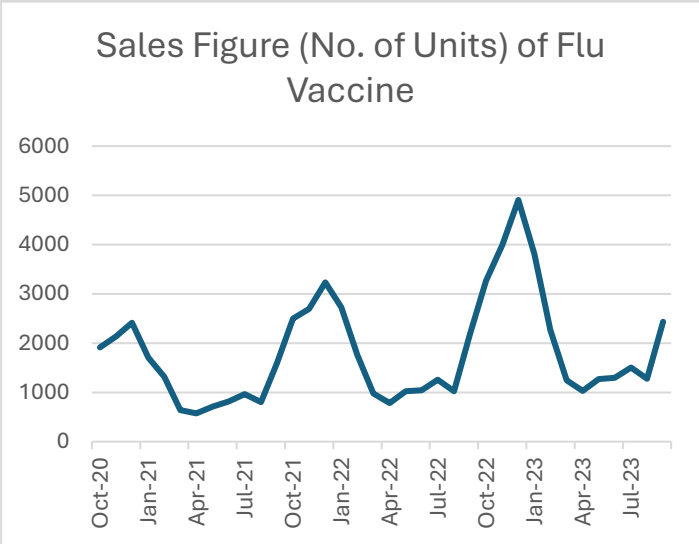
1. For each data set in the above Excel file, (Ashikeen)

   a) Plot a relevant graph



School of Engineering





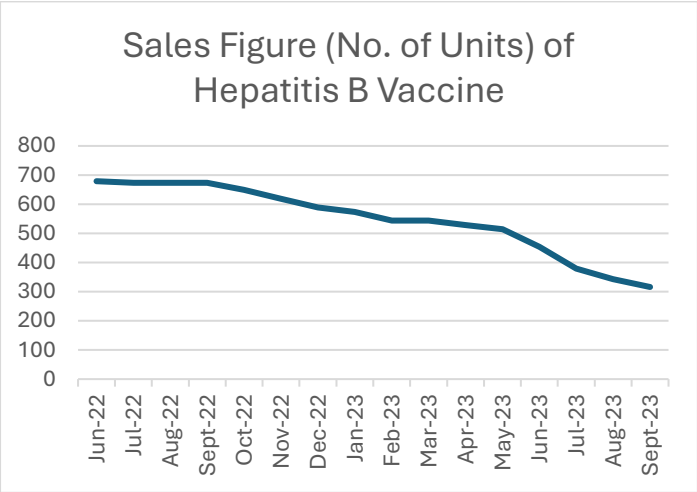
b) Identify the time series components present
   1. Random
   2. Random
   3. Trend
   4. Seasonality

c) Recommend a suitable forecasting method
   1. Moving Average
   2. Weighted Moving Average
   3. Holt's Method
   4. Multiplicative decomposition

Causal Forecasting – Simple Linear Regression

Which data set do you recommend using linear regression as the forecasting method?

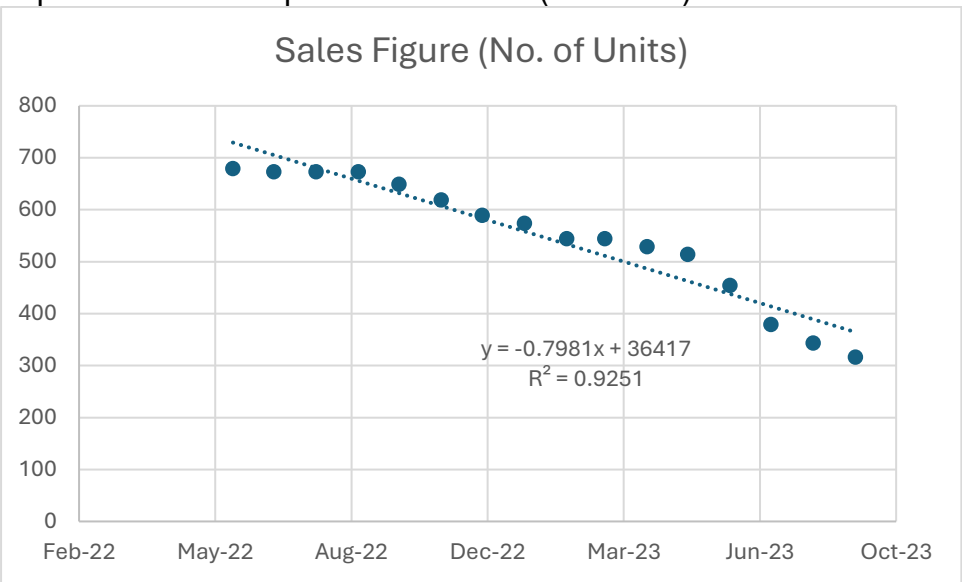
School of Engineering



2. For this data set, identify the dependent variable and the independent variable. (Ashikeen)

   Dependent Variable: Time
   Independent Variable: Sales figures

3. Use the scatter plot you have plotted to investigate the relationship between the dependent and independent variable. (Ashikeen)



   a) The relationship between the dependent variable and the independent variable? (Hint: linear or non-linear? Positive or negatively correlated?)
   There is a negative, linear relationship between time and the sales units of the Hepatitis B vaccine.

   b) Plot a linear regression line. The equation of the data's trend line is -0.9251.

   $\beta_0 = -0.7891$, $\beta_1 = 36417$

4. Obtain and interpret the following regression model statistics:
   a) Correlation coefficient, r -SQRT(0.9251) = -0.96182
   b) Coefficient of determination, R^2 ('R Square') 0.9251

5. Using the linear regression model, predict the dependent variable if the independent variable has a value of 2500.
   Dependent variable (y), Independent variable (x)
   Y = -0.7981x + 36417
   = -0.7981*(2500) + 36417
   = 34421.75

Time Series Forecasting – Moving Average and Exponential Smoothing [Celest]

Which data set do you recommend using moving average and exponential smoothing as the forecasting methods? MMR Vaccine

6. Moving Average is commonly used to derive forecast for time series with random/irregular variations. For the data set under study, develop a N-period moving average model that forecasts the monthly sales. Given N is the number of averaging periods,

   a) Use N = 3 (You can assume other values for N) to forecast the monthly sales.

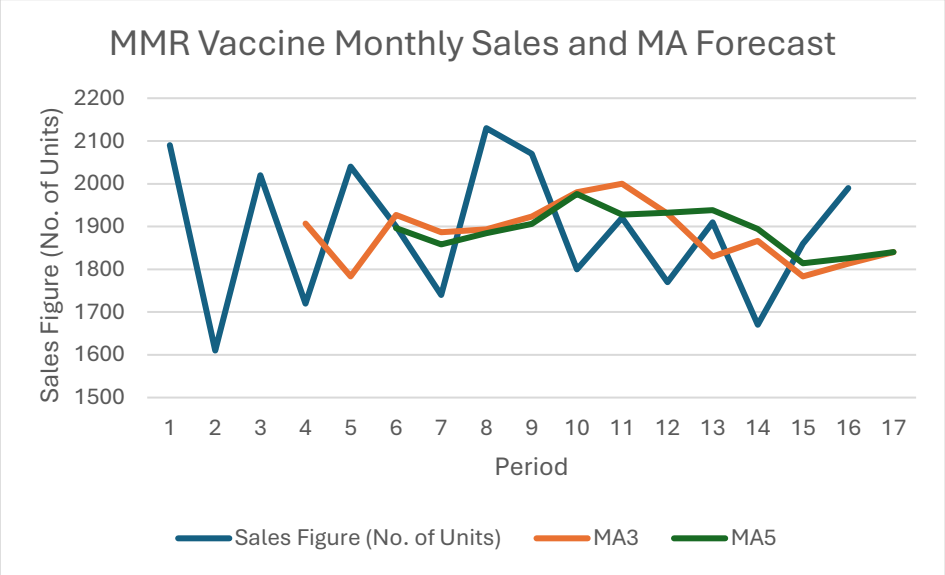
Month	Sales Figure (No. of Units)	MA3
Jun-22	2090	
Jul-22	1610	
Aug-22	2020	
Sep-22	1720	1906.67
Oct-22	2040	1783.33
Nov-22	1900	1926.67
Dec-22	1740	1886.67
Jan-23	2130	1893.33
Feb-23	2070	1923.33
Mar-23	1800	1980
Apr-23	1920	2000
May-23	1770	1930
Jun-23	1910	1830
Jul-23	1670	1866.67
Aug-23	1860	1783.33
Sep-23	1990	1813.33
		1840

   b) Use N = 5 (You can assume other values for N)) to forecast the monthly sales.

Month	Sales Figure (No. of Units)	MA3	MA5
Jun-22	2090		
Jul-22	1610		
Aug-22	2020		
Sep-22	1720	1906.67	
Oct-22	2040	1783.33	
Nov-22	1900	1926.67	1896
Dec-22	1740	1886.67	1858
Jan-23	2130	1893.33	1884
Feb-23	2070	1923.33	1906
Mar-23	1800	1980	1976
Apr-23	1920	2000	1928
May-23	1770	1930	1932
Jun-23	1910	1830	1938
Jul-23	1670	1866.67	1894
Aug-23	1860	1783.33	1814
Sep-23	1990	1813.33	1826
		1840	1840

c) Calculate Mean Absolute Deviation (MAD) and Mean Absolute Percent Error (MAPE) for a) and b).
MA3 → MAD = 150.00, MAPE = 7.97%
MA5 → MAD = 121.82, MAPE = 6.49%

d) Plot the forecasted values from a) and b) with the actual sales figures and comment on the graphs.



N=3 MA graph is more responsive to changes in sales figures compared with N=5 MA graph.

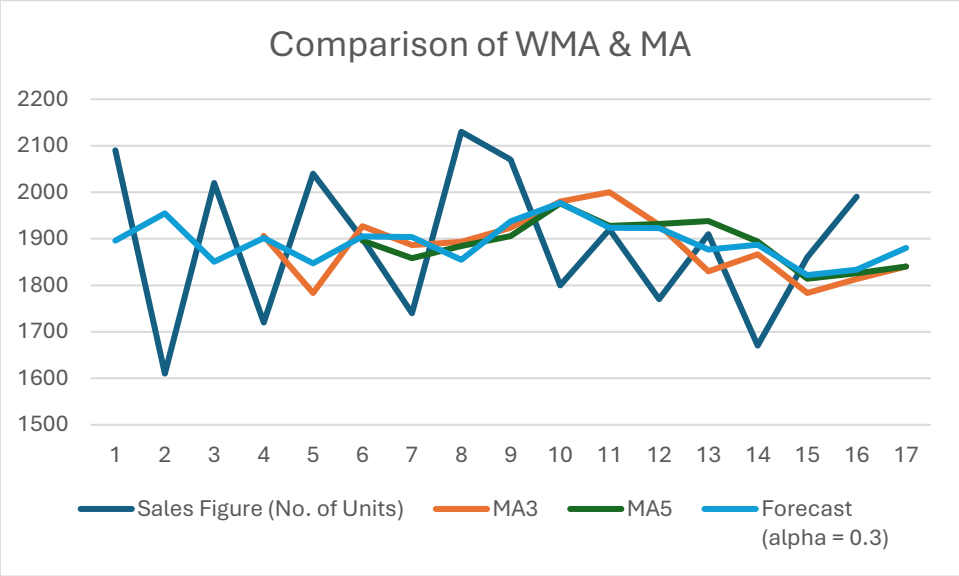
e) What happens to the forecast when you increase N?
As N increases, the forecast becomes less responsive to changes in actual data and the graph becomes smoother.

7. For the forecasts in Q6, the underlying assumption is that all data points are of equal importance (equal weights).

a) Assign weights (You can assume values for the weights) to the 3-period moving average to predict the monthly sales.
Weight 5,3,1

b) Calculate Mean Absolute Deviation (MAD) and Mean Absolute Percent Error (MAPE) for the weighted moving average forecasts.
MAD = 151.88, MAPE = 0.08

c) Plot the forecasted values to the moving average graphs and compare the use of **weighted moving average** and **moving average** in forecasting.

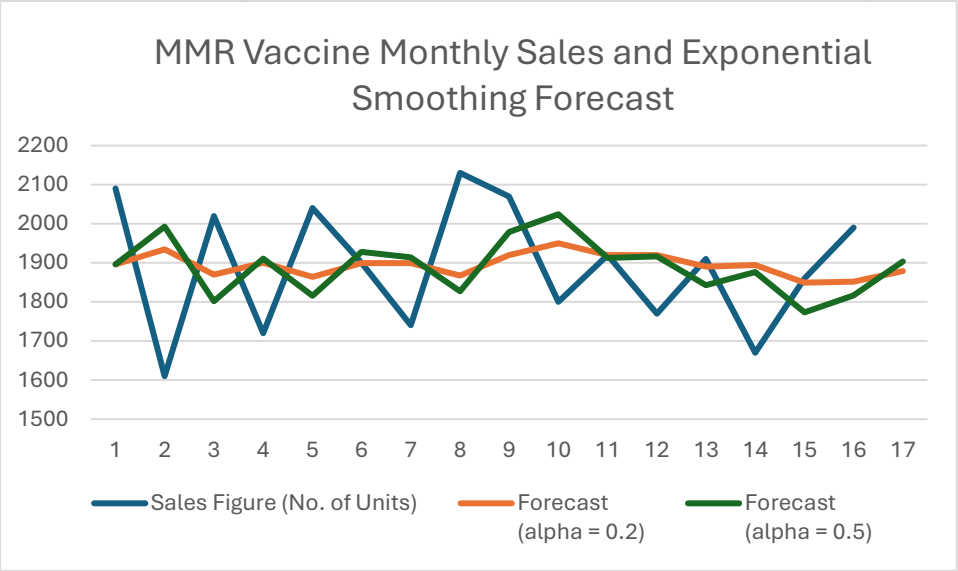


WMA is more responsive to current trends compared with moving average, which weighs all points equally.

8. Another commonly used forecasting method for time series with random/irregular variations is simple exponential smoothing. Develop a **simple exponential smoothing** model for the same data set to forecast the monthly sales. Given that α (Alpha) is the smoothing constant,

a) Use α = 0.2 (You can assume other values for α) to predict the monthly sales.
b) Use α = 0.5 (You can assume other values for α) to predict the monthly sales.
c) Calculate Mean Absolute Deviation (MAD) and Mean Absolute Percent Error (MAPE) for a) and b).
MAD (when α = 0.2) = 143.20, MAPE = 7.72%
MAD (when α = 0.5) = 169.92, MAPE = 9.15%

d) Plot the forecasted values from a) and b) with the actual sales figures and comment on the graphs. What happens to the forecast when you vary *α*?



α =0.5 SES is more responsive to changes in actual sales compared with α =0.2 SES. As α decreases, the forecast becomes less responsive to changes in actual data and the graph becomes smoother.

9. Tabulate and compare the forecasting errors for the methods applied in Q6 to Q8. Which forecasting method is most suitable for forecasting sales in Oct-23?

Forecasting methods	**Measures of forecasting error**	
	MAD	MAPE
MA (3-Period)	150.00	7.97%
MA (5-Period)	121.82	6.49%
WMA (3-Period)	151.88	8.08%
SES (0.2)	143.20	7.72%
SES (0.5)	169.92	9.15%

Time Series Forecasting – Trend Analysis using Linear Regression and Holt's Method

Which data set do you recommend using Holt's method as the forecasting method? MMR Vaccine-DisExp

10. Perform Linear Regression to forecast the monthly sales of this data set.

School of Engineering

x (independent variable)	y (dependent variable)		
Distribution Expenses ($)	Sales Figure (No. of Units)	Forecasted Sales Prices	AD
4682	2090	2121.107	31.107
3820	1610	1704.33	94.33
4496	2020	2031.176	11.176
3793	1720	1691.2755	28.7245
4367	2040	1968.8045	71.1955
4375	1900	1972.6725	72.6725
4028	1740	1804.898	64.898
4504	2130	2035.044	94.956
4548	2070	2056.318	13.682
4015	1800	1798.6125	1.3875
4276	1920	1924.806	4.806
3837	1770	1712.5495	57.4505
4237	1910	1905.9495	4.0505
3666	1670	1629.871	40.129
4156	1860	1866.786	6.786
4468	1990	2017.638	27.638
		MAD	39.0618125

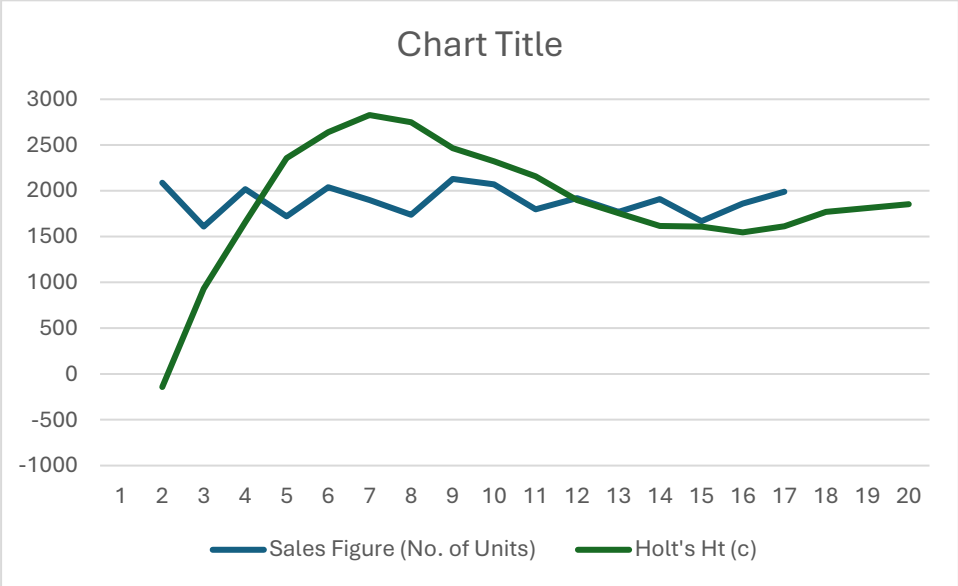
$y = 0.4835x – 142.64$

Sub x (distribution expenses) to get the forecasted sales prices respectively.

11. Based on Holt's method, determine the forecasts up to Dec-23. Assume α (for level estimation) = 0.3 and β (for trend estimation) = 0.6 (You can assume other values for the smoothing constants α and β).

    a) What values would you use from your regression analysis to represent L0 and T0?
    L0 = -142.6375, T0 = 0.4835

    b) Plot the forecasted values with the actual sales figures and comment on the graphs.



12. Compare the MAD and MAPE of Holt's method and Linear Regression forecasts. Which method would you recommend deriving forecasts for the monthly sales of this data set? Explain.
    I will recommend Holt's method as it gives a more accurate forecast due to having smaller forecasting errors.

    Holt's method:

MAD	529.5241	
MAPE		27.79%

    Linear Regression:

MAD	2028.53	
MAPE		107.38%

Time Series Forecasting – Handling Trend and Seasonality using Decomposition Method

Which data set do you recommend using Multiplicative Decomposition Method as the forecasting method? Flu Vaccine

15. There are two general forms of time-series decomposition models: multiplicative and additive. Which model is more suitable to be used to forecast the monthly sales of the data set? Justify your answer.
    Multiplicative model is more suitable to be used to forecast the monthly sales. As the seasonal trend increases every cycle rather than staying constant.

16. Write down the decomposition model identified in Q15 that Andy can use to forecast the monthly sales based on the four time series components.

    Let Y_t be the observed value of the time series in time t;
    $\mathbf{T}_t$ be the trend component in time t;
    $\mathbf{S}_t$ be the seasonal component in time t;
    $\mathbf{C}_t$ be the cyclic component in time t; and
    $\mathbf{I}_t$ be the irregular (or random) component in time t.

17. In the decomposition method, Moving Averages (MA) and/or Centred Moving Averages (CMA) are often used to eliminate seasonal variations and irregular fluctuations from the data. Answer the following questions:

    a) To remove the seasonal effect, what value should <u>the number of averaging periods</u> (N) be in MA calculation?
    N = 12

    b) Write down the formula to calculate the MA at Period 6 (Mar-21). Calculate the rest of the moving averages for this data set.

School of Engineering

REPUBLIC POLYTECHNIC

Mar-21 = AVERAGE(B3:B14)

Rest of MA for this data set:

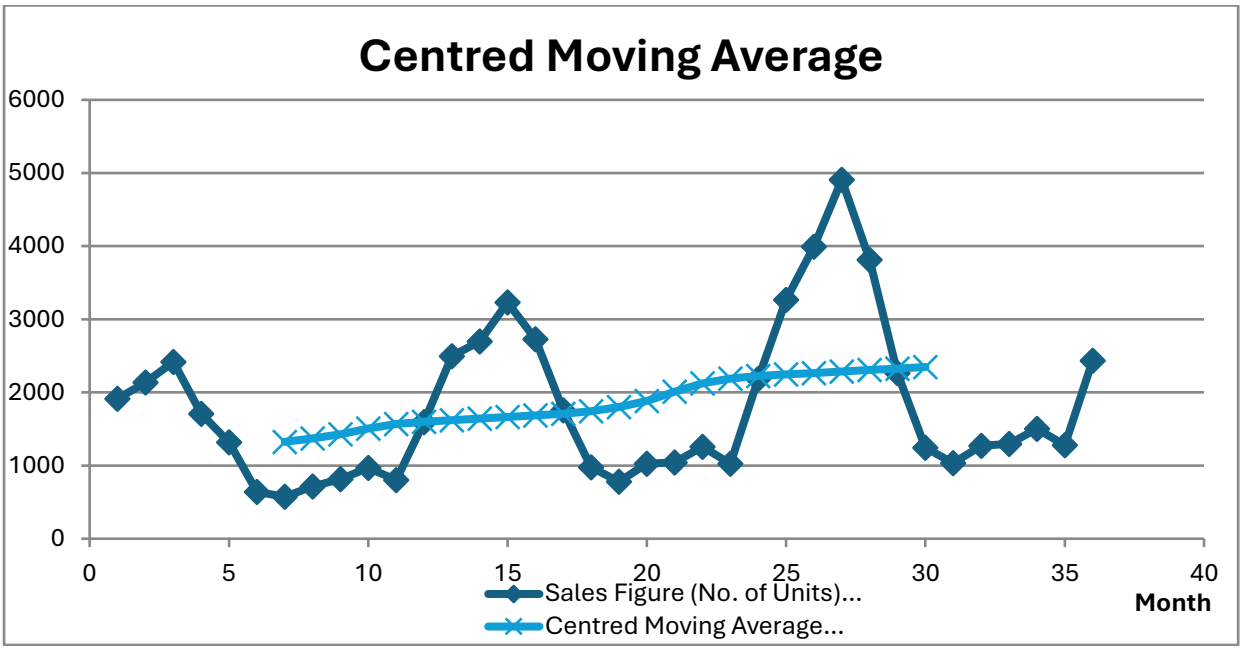
Month	Sales Figure (No. of Units) (Y)	Time Period (X)	Moving Average (e.g. 12 months)
Oct-20	1915	1	
Nov-20	2136	2	
Dec-20	2417	3	
Jan-21	1709	4	
Feb-21	1319	5	
Mar-21	639	6	1300.417
Apr-21	574	7	1348.833
May-21	713	8	1395.500
Jun-21	816	9	1463.333
Jul-21	967	10	1548.167
Aug-21	804	11	1584.500
Sept-21	1596	12	1612.667
Oct-21	2496	13	1629.917
Nov-21	2696	14	1656.000
Dec-21	3231	15	1675.083
Jan-22	2727	16	1699.250
Feb-22	1755	17	1717.667
Mar-22	977	18	1766.833
Apr-22	781	19	1831.000
May-22	1026	20	1939.167
Jun-22	1045	21	2078.833
Jul-22	1257	22	2169.250
Aug-22	1025	23	2211.833
Sept-22	2186	24	2234.000
Oct-22	3266	25	2255.000
Nov-22	3994	26	2275.417
Dec-22	4907	27	2296.250
Jan-23	3812	28	2316.667
Feb-23	2266	29	2337.833
Mar-23	1243	30	2358.750
Apr-23	1033	31	
May-23	1271	32	
Jun-23	1295	33	
Jul-23	1502	34	
Aug-23	1279	35	
Sept-23	2437	36	

c) When and why do you need to calculate the CMA? How is it different from MA?
CMA is calculated only when N is an even number to remove the seasonal variations from data.

d) Calculate the CMA for Period 7 (Apr-21) onwards.

Month	Sales Figure (No. of Units) (Y)	Time Period (X)	Moving Average (e.g. 12 months)	Centred Moving Average (=(MA_{i-1} + MA_i)/2)
Apr-21	574	7	1348.833	1324.625
May-21	713	8	1395.500	1372.167
Jun-21	816	9	1463.333	1429.417
Jul-21	967	10	1548.167	1505.750
Aug-21	804	11	1584.500	1566.333
Sept-21	1596	12	1612.667	1598.583
Oct-21	2496	13	1629.917	1621.292
Nov-21	2696	14	1656.000	1642.958
Dec-21	3231	15	1675.083	1665.542
Jan-22	2727	16	1699.250	1687.167
Feb-22	1755	17	1717.667	1708.458
Mar-22	977	18	1766.833	1742.250
Apr-22	781	19	1831.000	1798.917
May-22	1026	20	1939.167	1885.083
Jun-22	1045	21	2078.833	2009.000
Jul-22	1257	22	2169.250	2124.042
Aug-22	1025	23	2211.833	2190.542
Sept-22	2186	24	2234.000	2222.917
Oct-22	3266	25	2255.000	2244.500
Nov-22	3994	26	2275.417	2265.208
Dec-22	4907	27	2296.250	2285.833
Jan-23	3812	28	2316.667	2306.458
Feb-23	2266	29	2337.833	2327.250
Mar-23	1243	30	2358.750	2348.292

18. Plot a graph for the CMA and the actual values, what do you observe?



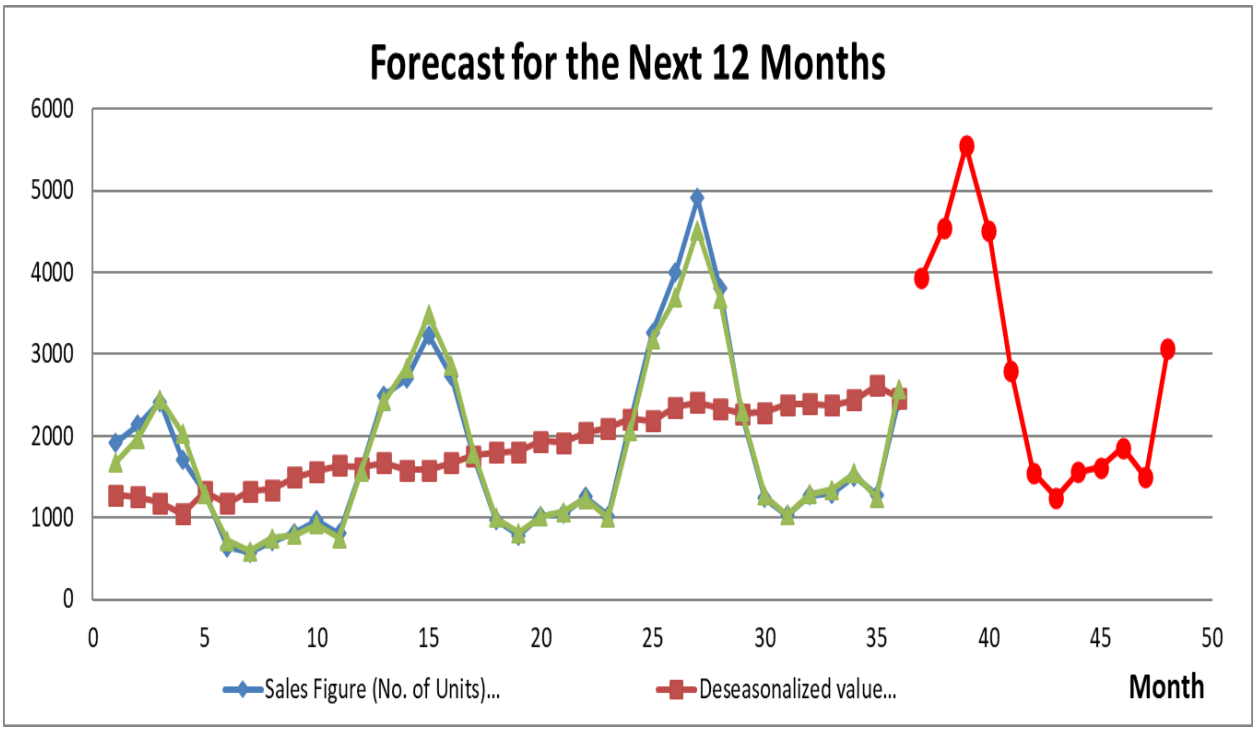
19. Produce the forecasts for the next 12 months from Oct 2023 by working out the following in order:

    a) Seasonal ratio
    b) Seasonal index
    c) Deseasonalized value
    d) Deseasonalized forecast
    e) Seasonalized forecast

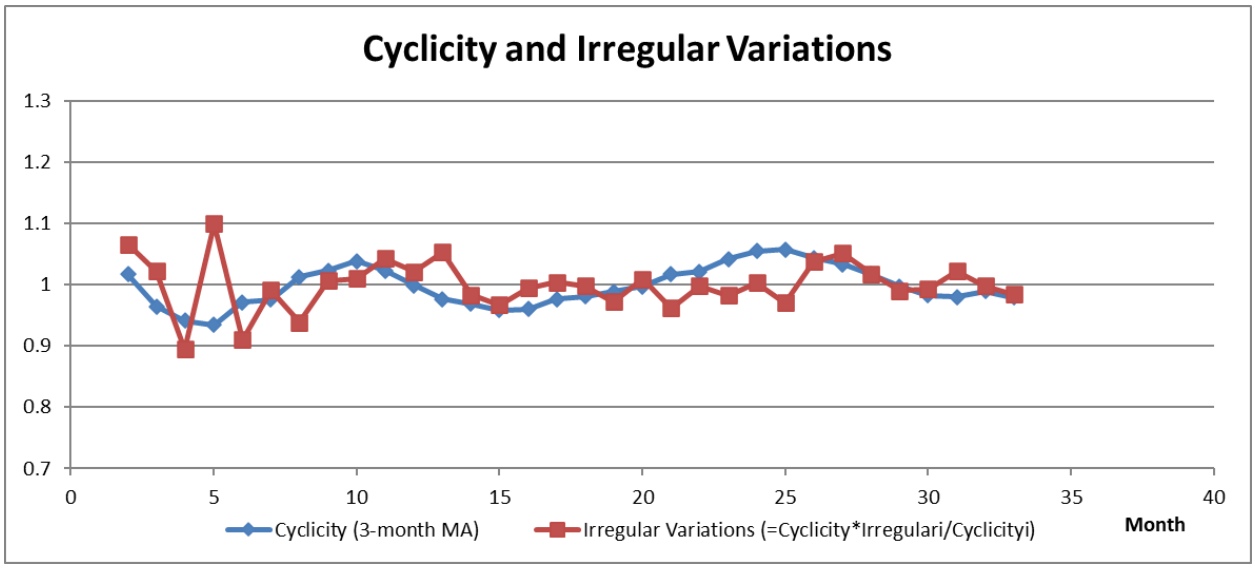
School of Engineering

Seasonal Ratio (=Y_i/CMA_i)	Seasonal Index (= normalized average of seasonal ratio)	Deseasonalized value (= Y_i / Seasonal $Index_i$)	Deseasonalized forecast (=slope*X_i + intercept)	Seasonalized forecast (= Seasonal $Index_i$ *Deseasonalized $forecast_i$)
	1.493	1282.413	1118.658	1670.467
	1.697	1258.335	1160.905	1970.614
	2.038	1186.084	1203.151	2451.781
	1.630	1048.382	1245.398	2030.161
	0.998	1321.954	1287.645	1284.768
	0.544	1175.548	1329.891	722.898
0.433	0.433	1326.948	1372.138	593.548
0.520	0.531	1343.987	1414.385	750.347
0.571	0.544	1499.887	1456.632	792.467
0.642	0.615	1571.494	1498.878	922.317
0.513	0.489	1643.201	1541.125	754.055
0.998	0.988	1615.028	1583.372	1564.717
1.540	1.493	1671.490	1625.618	2427.501
1.641	1.697	1588.235	1667.865	2831.170
1.940	2.038	1585.534	1710.112	3484.864
1.616	1.630	1672.872	1752.359	2856.573
1.027	0.998	1758.930	1794.605	1790.596
0.561	0.544	1797.356	1836.852	998.469
0.434	0.433	1805.482	1879.099	812.844
0.544	0.531	1933.984	1921.345	1019.295
0.520	0.544	1920.812	1963.592	1068.274
0.592	0.615	2042.780	2005.839	1234.269
0.468	0.489	2094.878	2048.086	1002.105
0.983	0.988	2212.062	2090.332	2065.705
1.455	1.493	2187.134	2132.579	3184.534
1.763	1.697	2352.898	2174.826	3691.726
2.147	2.038	2407.990	2217.072	4517.948
1.653	1.630	2338.463	2259.319	3682.985
0.974	0.998	2271.074	2301.566	2296.424
0.529	0.544	2286.707	2343.813	1274.041
	0.433	2388.045	2386.059	1032.141
	0.531	2395.802	2428.306	1288.244
	0.544	2380.336	2470.553	1344.082
	0.615	2440.935	2512.799	1546.221
	0.489	2613.998	2555.046	1250.155
	0.988	2466.054	2597.293	2566.692

20. Plot the monthly sales forecasts for the next 12 months. Comment on the validity of the Multiplicative Decomposition Model built.



21. Investigate if there are significant cyclical components in the sales data by further decomposing the seasonal forecasts into cyclical and irregular components.
[Copilot]
There are no significant cyclical components in the sales data.
After removing trend and seasonal effects, the remaining cyclical–irregular series fluctuates randomly around 1.0 without any stable or repeating multi-year pattern. This indicates that the deviations are primarily due to irregular (random) variation rather than true cyclical behaviour.

22. Plot the cyclical and irregular components of the time series. Do they have significant effect on the forecasts? If under certain conditions, the cyclical component plays a significant role in the forecast, how would you incorporate it into the forecasting?

School of Engineering